

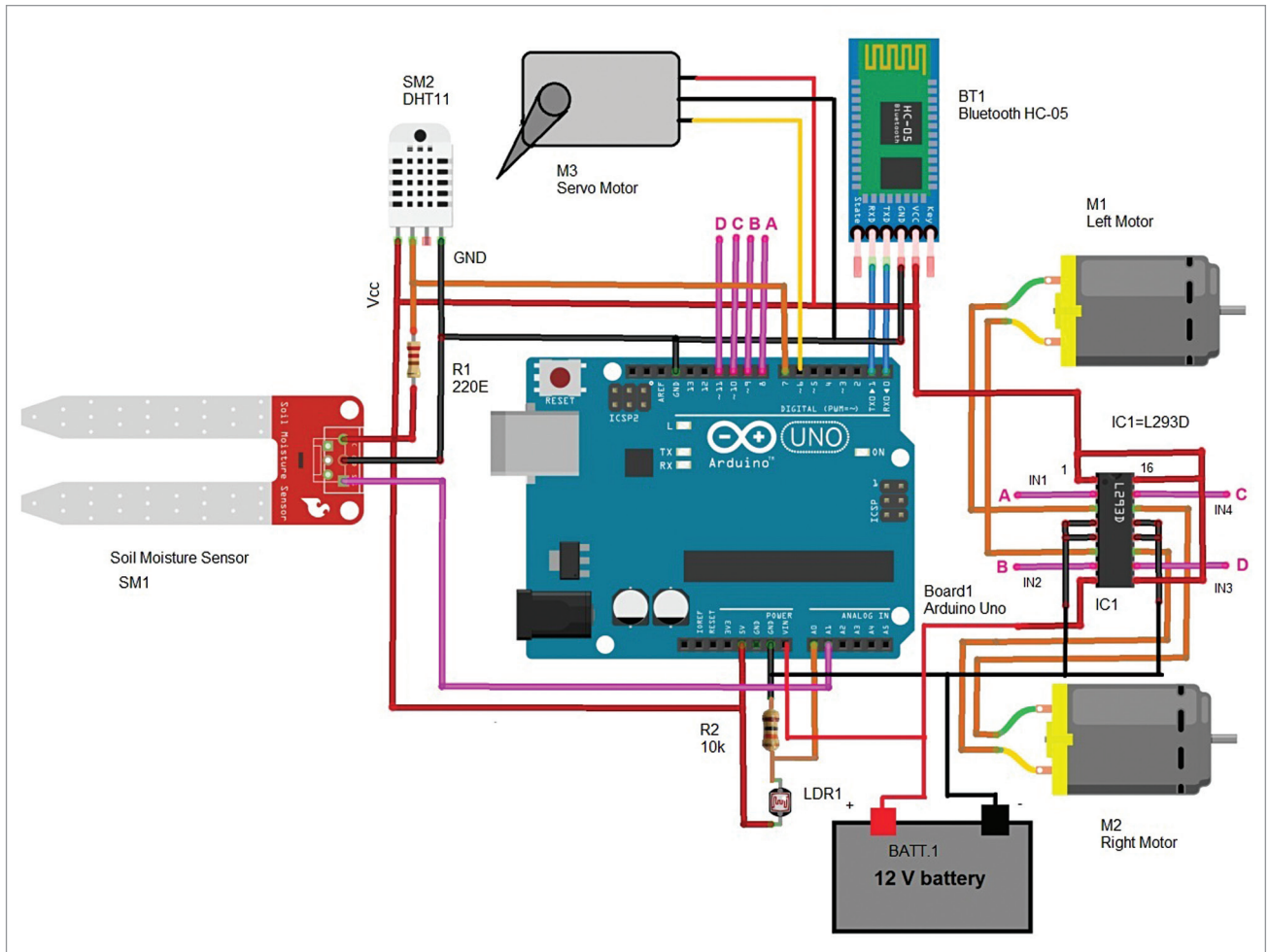


Fig. 2: Circuit diagram

all three sensors to Android smart-phone through Bluetooth module

Motor driver L293D (IC1) provides sufficient voltage and current to both the motors to rotate them. It amplifies the output of Arduino board (Board1) and drives the motors.

The DC motors (M1 and M2) drive left and right wheels of the robot and move it forward, backward, left, and right.

Soil moisture sensor (SM1) is attached to servo motor (M3) shaft. This motor moves the sensor up and down to insert it into the soil to check soil moisture content.

HC-05 module operates on 5V received from the Arduino board. It communicates with Arduino board with USART pins Tx (D1) – Rx (D0).

So, its Tx pin is connected to Rx pin of Arduino board and vice versa.

DHT11 sensor also gets its 5V supply from Arduino board. Its digital output is connected to digital pin D7 of Arduino.

The analogue output of soil moisture sensor (SM1) is connected to analogue input pin A1 of Arduino board. Its 5V supply also comes from Arduino board.

LDR1 is configured in pulled-down mode with 10-kilo-ohm pull-down resistor. Its analogue output is given to analogue input pin A0 of Arduino board.

Digital pins D8, D9, D10, D11 of Arduino board drive DC motors M1 and M2 using L293D chip. These pins are connected to inputs of L293D, and two motors are con-

nected to output of the chip. Servo motor signal (sig) input is connected to PWM output pin D6 of Arduino board. The motor gets 5V supply from Arduino board.

The motor supply pin Vss of L293D (pin 8) gets 12V from battery. The Arduino board also gets 12V input at its Vin pin from battery. That is, Vin pin gets 12V input and gives 5V output to all other components.

Circuit and operation

The circuit operation starts when 12V battery is connected to Arduino Uno board and L293D chip. Initially, both motors (M1 and M2) are at rest and so the robot is also at rest. The servo motor (M3) is at 0° position and soil moisture sensor is in upward position.